

BVAR vs. Information Criteria: A Monte Carlo Simulation Study

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Abstract

This is the abstract for your paper. Briefly state the purpose of the research, the principal results, and major conclusions. The abstract should be self-contained and citation-free.

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1 Introduction

Introduce the problem of lag selection and parameter proliferation in Structural VARs. Highlight the debate between hard lag selection (Information Criteria) and Bayesian methods (Shrinkage and Model Averaging).

2 Methodology

2.1 Data Generating Process

Explain the true DGP based on Kilian and Murphy (2014) or other literature.

2.2 Optimal Prior Selection via Marginal Data Density

Describe how the shrinkage parameter (τ) is treated as an optimization problem rather than a fixed heuristic.

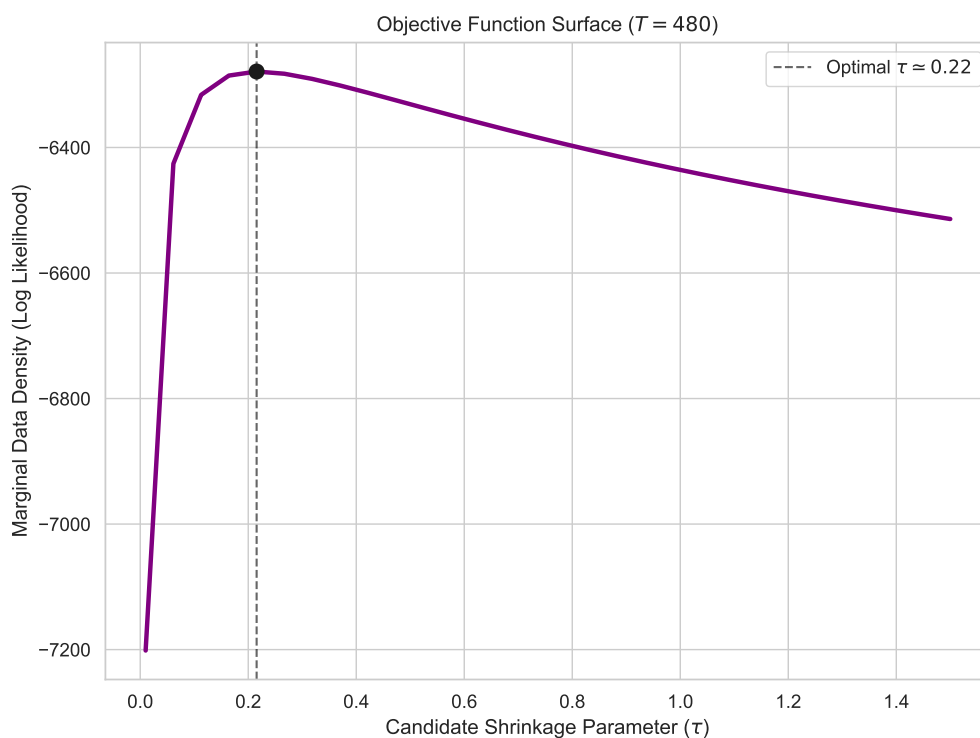


Figure 1: Objective Function Surface for Marginal Data Density Optimization. The peak represents the optimal balance between model fit and parameter penalty.

3 Simulation Results

Present the results of the Monte Carlo simulations, broken down by the specific strategies employed to handle model uncertainty.

3.1 Behavior of Standard Information Criteria

Before evaluating the Bayesian estimators, we examine the baseline lag detection performance of the classical Information Criteria.

Table 1: Average Evaluated Lag Order ($\hat{p}$) selected by Information Criteria.

	Estimator	AIC	SIC/BIC	HQC
p_0	T			
4	96	10.30	1.30	1.80
	144	2.60	1.40	2.10
	240	3.10	2.00	2.10
	480	3.80	2.00	2.40
6	96	11.50	1.30	1.90
	144	3.40	1.50	2.00
	240	4.20	2.00	2.10
	480	5.80	2.10	2.50
8	96	12.00	1.40	3.00
	144	7.60	1.50	2.00
	240	6.50	1.90	2.00
	480	8.00	2.00	2.10
10	96	10.20	1.20	1.70
	144	5.00	1.50	2.00
	240	5.70	2.00	2.10
	480	9.40	2.00	2.50

3.2 The Uncertainty Strategy: Averaging vs. Selection

Here, we evaluate True Bayesian Model Averaging (BMA) against Information Criteria. We focus on how BMA acts as an insurance policy against catastrophic lag misspecification in small samples.

Table 2: Relative Mean Squared Error: Hard Selection vs. Bayesian Model Averaging.

T	p_0	AIC	SIC/BIC	HQC	OLS BMA (BIC-W)	BMA BVAR-WN (BIC-W, $\tau = 0.20$)	BMA BVAR-WN (BIC-W, $\tau = 0.50$)
96	4	1.921	0.958	1.253	1.234	1.095	1.248
	6	1.229	1.248	1.089	0.893	0.822	1.039
	8	1.813	1.298	1.287	1.113	1.170	1.096
	10	1.179	0.813	0.604	0.617	0.698	0.538
144	4	1.330	1.363	1.252	1.787	1.231	1.484
	6	1.181	0.689	0.756	0.983	0.686	0.676
	8	0.672	1.333	0.755	0.841	1.072	0.903
	10	0.556	0.824	0.793	1.050	0.299	0.637
240	4	0.701	0.589	0.685	0.912	0.743	0.840
	6	1.011	1.046	1.111	0.789	0.944	0.817
	8	1.177	1.372	1.416	1.095	1.006	0.742
	10	1.071	1.132	1.307	0.981	1.154	1.636
480	4	1.091	2.121	2.015	1.373	2.100	1.435
	6	0.688	0.762	0.635	1.149	0.924	1.520
	8	1.000	0.914	0.826	0.740	0.527	0.755
	10	1.340	1.042	1.005	0.994	1.671	0.787

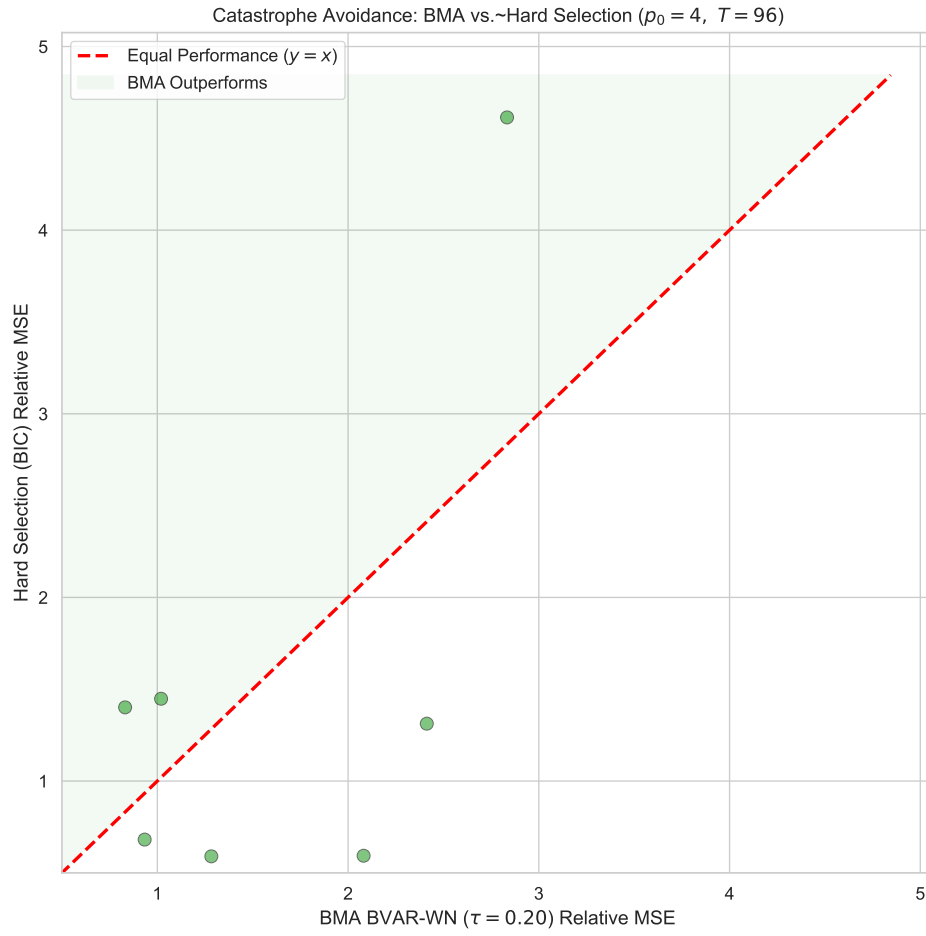


Figure 2: Catastrophe Avoidance: BMA vs. Hard Selection (BIC). Points above the 45-degree line indicate iterations where BMA outperformed BIC, notably smoothing out catastrophic tail risks.

Table 3: Posterior Lag-Weight Distribution for BMA ($p_0 = 4$). OLS BMA and BMA BVAR-WN share identical BIC-based lag weights.

Estimator	Hybrid-BMA (OLS W)			
T	96	144	240	480
Statistic / Lag				
$E[p]$	4.11	4.44	5.04	5.68
p=1	19.7%	16.6%	12.8%	10.0%
p=2	18.5%	17.0%	14.1%	11.5%
p=3	14.6%	14.1%	12.7%	11.0%
p=4	11.8%	11.4%	11.3%	10.3%
p=5	8.9%	9.4%	9.7%	9.4%
p=6	6.6%	7.6%	8.4%	8.7%
p=7	5.2%	6.2%	7.2%	8.0%
p=8	4.1%	5.0%	6.2%	7.3%
p=9	3.2%	4.1%	5.4%	6.7%
p=10	2.6%	3.4%	4.7%	6.2%
p=11	2.5%	2.8%	4.0%	5.7%
p=12	2.3%	2.3%	3.5%	5.2%

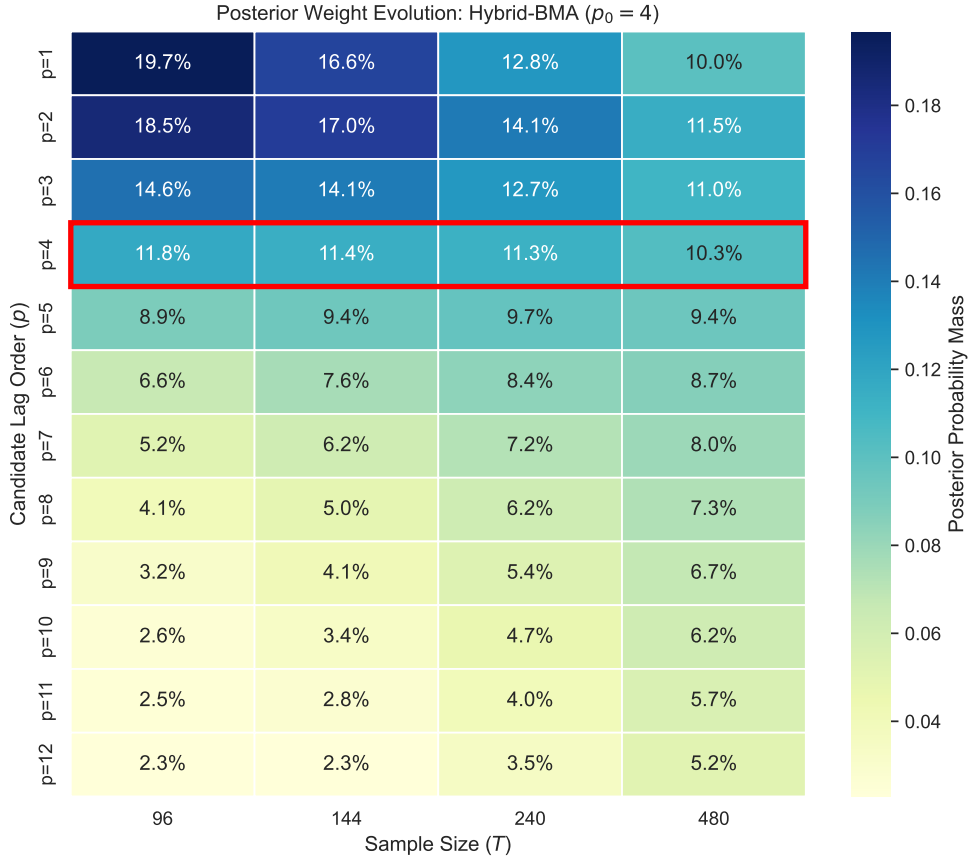


Figure 3: Posterior Weight Evolution of SOTA-BMA. At small sample sizes, probability mass is dispersed to hedge uncertainty. As T grows, the distribution concentrates precisely on the true DGP.

4 Conclusion

Summarize the main findings, specifically highlighting how optimal τ selection and true BMA systematically outperform standard econometric heuristics in finite samples.